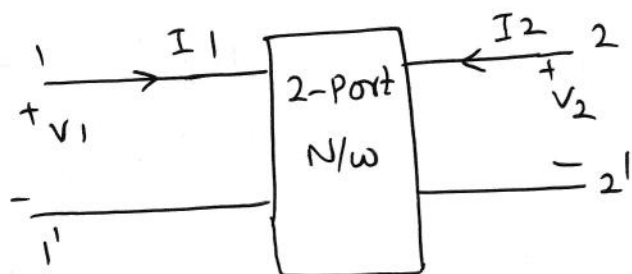


* Two Port Networks:

* Two Ports:

→ It represents two pairs of terminals describing a n/w where the +ve reference voltage polarities and current directions are as shown below.



→ Out of the 4 parameters V_1, V_2, I_1, I_2 , 2 of them are considered as dependent and the other two as independent. So, we have $4C_2 = 6$ types of Parameters describing the network. They are:

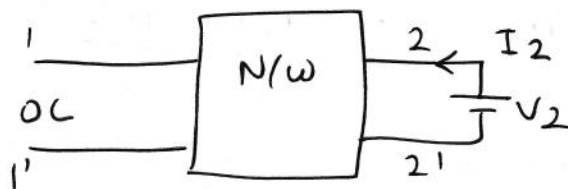
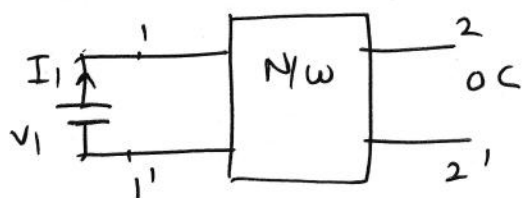
1) Z 2) Y 3) h 4) $ABCD$ 5) g & 6) $A'B'C'D'$

→ The network inside the ports consists of linear, passive, bilateral elements. They may also have energy storage elements, but their initial conditions must be zero.

→ The network inside may also have dependent sources, but never an independent source in it.

* Concept of symmetry:

If the ratio of excitation to response remains constant at both the ports independently w.r. to defined circuit condition such as open circuit or short circuit, then the network is said to be symmetrical.



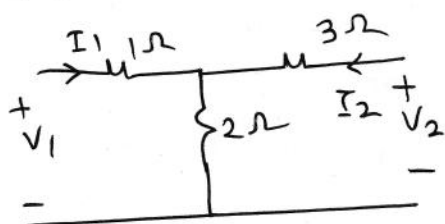
$$\boxed{\frac{V_1}{I_1} = \frac{V_2}{I_2}} \rightarrow \text{Symmetrical}$$

→ However for simple networks, symmetry can be identified as a mirror image property w.r.to ports.

* Concept of Reciprocity:

If it obeys reciprocity theorem, then it is a reciprocal n/w. However, any linear, passive, bilateral n/w is always reciprocal. ($\because$ Reciprocity means passivity). Reciprocity $\Leftrightarrow$ Passivity.

Q:- Det. Z Parameters for



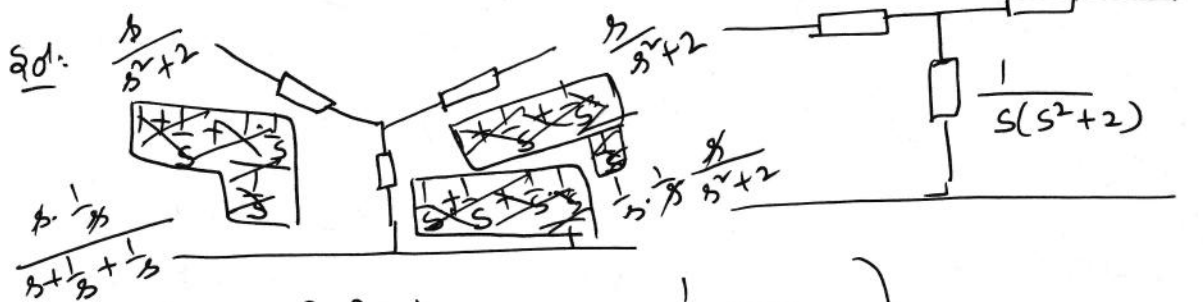
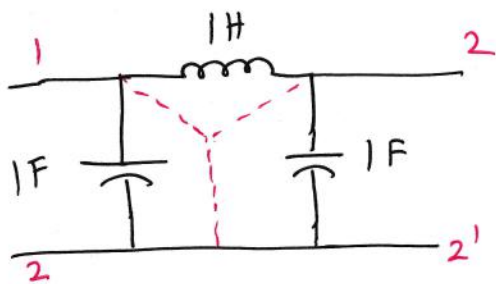
Sol. $Z_{11} = \begin{bmatrix} 3 & 2 \\ 2 & 5 \end{bmatrix}$

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

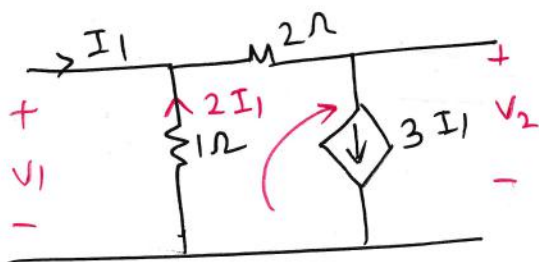
$$Z_{11} = \frac{V_1}{I_1} / I_2 = 0 \quad Z_{21} = \frac{V_2}{I_1} / I_2 = 0$$

Q: Det- Z Parameters for this



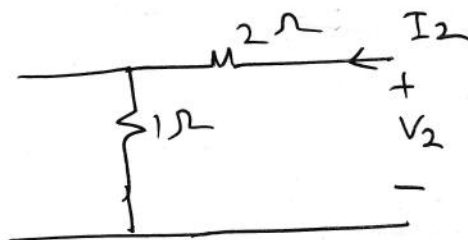
$$[Z] = \begin{pmatrix} \frac{s^2+1}{s(s^2+2)} & \frac{1}{s(s^2+2)} \\ \frac{1}{s(s^2+2)} & \frac{s^2+1}{s(s^2+2)} \end{pmatrix}$$

GATE



Sol:

$$\begin{aligned} V_1 &= -2I_1 \Rightarrow Z_{11} = -2\Omega \\ 2I_1 + 6I_1 + V_2 &= 0 \\ V_2 &= -8I_1 \\ \Rightarrow Z_{21} &= -8\Omega \end{aligned}$$

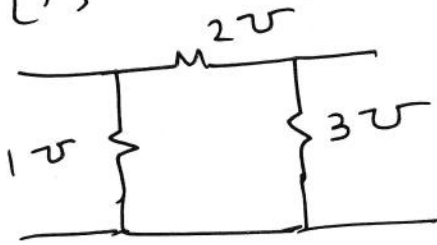


$$V_2 = 3I_2 \Rightarrow Z_{22} = 3\Omega$$

$$V_1 = I_2 \Rightarrow Z_{12} = \underline{\underline{1\Omega}}$$

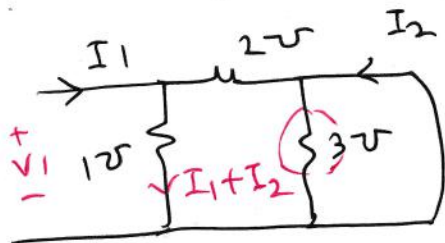
$$[Z] = \begin{pmatrix} -2 & 1 \\ -8 & 3 \end{pmatrix} \Omega$$

Q: [4]



Sol:

$$[Y] = \begin{bmatrix} 1+2 & -2 \\ -2 & 3+2 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -2 & 5 \end{bmatrix} \text{ S}$$



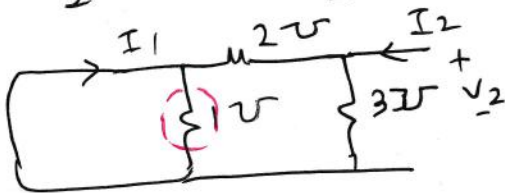
$$V_1 = \frac{I_1 + I_2}{1}$$

$$I_2/2 + \frac{I_1 + I_2}{1} = 0$$

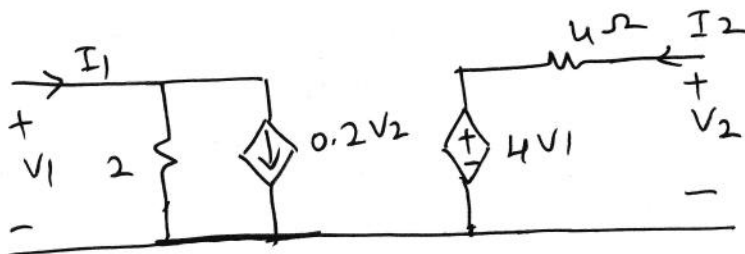
$$\Rightarrow I_1 = -\frac{3}{2} I_2$$

$$\therefore V_1 = I_1 - \frac{2}{3} I_1 = \frac{1}{3} I_1 \Rightarrow Y_{11} = \underline{3 \text{ S}}$$

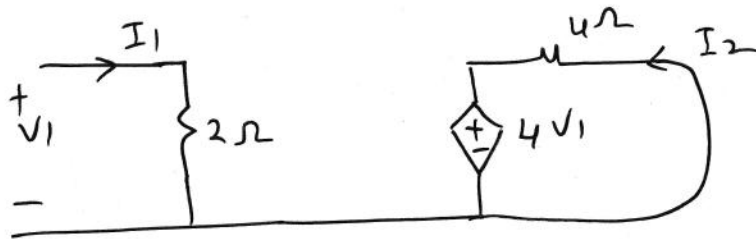
$$V_1 = -\frac{3}{2} I_2 + I_2 = -\frac{I_2}{2} \Rightarrow Y_{21} = \underline{-2 \text{ S}}$$



GATE
Q:



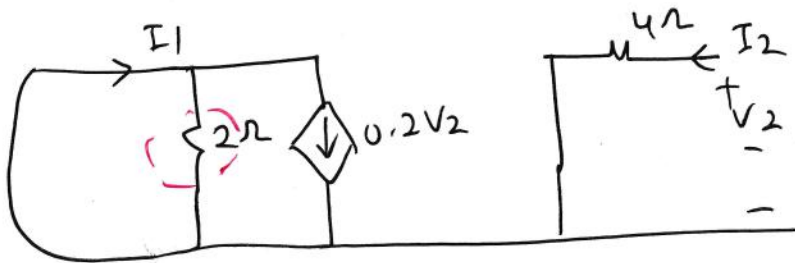
sol.



$$V_1 = 2 I_1 \rightarrow y_{11} = \frac{1}{2}$$

$$4 I_2 + 4 V_1 = 0$$

$$V_1 = -I_2 \Rightarrow y_{21} = -1 \Omega$$



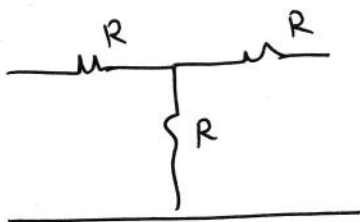
$$V_2 = 4 I_2 \Rightarrow y_{22} = \frac{1}{4} \Omega$$

$$I_1 = 0.2 V_2 \Rightarrow y_{12} = 0.2 \Omega$$

$$\therefore [Y] = \begin{bmatrix} 1/2 & 0.2 \\ -1 & 1/4 \end{bmatrix} \Omega$$

GATE 1M:

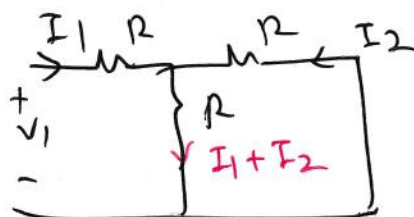
Find $h_{21} = \underline{\hspace{2cm}}$



$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$h_{21} = \frac{I_2}{I_1} \bigg|_{V_2=0}$$



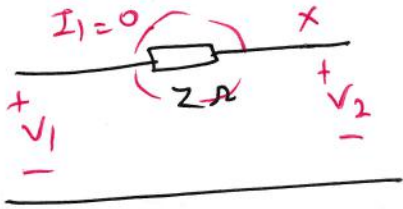
$$[I_2 + (I_1 + I_2)] R = 0$$

$$I_1 = -2 I_2$$

$$\therefore \frac{I_2}{I_1} = -\frac{1}{2}$$

Q: Det. the ABCD parameters for series branch.

[T]



$$V_1 = AV_2 + BI_2$$

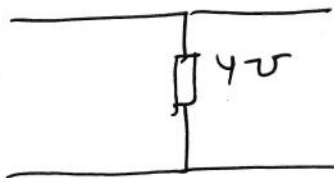
$$I_1 = CV_2 + DI_2$$

$$A = \frac{V_1}{V_2} \bigg|_{I_2=0} = 1.$$

$$C = \frac{I_1}{V_2} = 0.$$

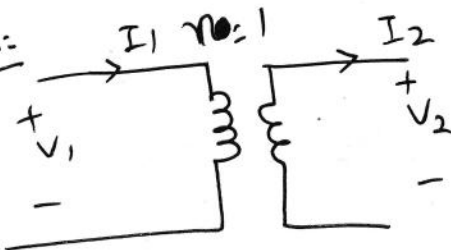
$$\begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$$

Q: [T]



$$\begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix}$$

GATE:



$$V_1 = AV_2 + BI_2$$

$$I_1 = CV_2 + DI_2$$

M/C model

$$k = \frac{n_2}{n_1} = \frac{1}{n} = \frac{V_2}{V_1} = \frac{I_1}{I_2}$$

$$V_1 = (n)V_2 + (0)I_2$$

$$I_1 = (0)V_2 + \left[\frac{1}{n}\right]I_2$$

$$\begin{bmatrix} n & 0 \\ 0 & 1/n \end{bmatrix}$$

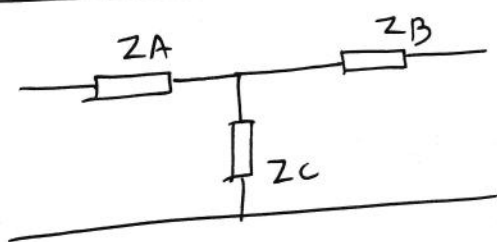
$$AD - BC = n \cdot \frac{1}{n} - 0 = 1$$

→ Reciprocal

→ Two port n/ws allows us to model complex practical n/w and systems into their simple equivalents in terms of their port parameter values.

→ Here, it is our basic assumption to model such big n/ws into standard canonical forms such as T, π , Lattice, bridged T, T-T, Ladder etc.

1) T-equivalent representation of a 2-port reciprocal n/w in terms of Z-parameters.

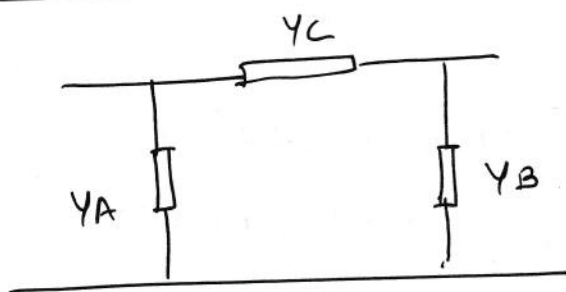


$$Z_A = Z_{11} - Z_{12}$$

$$Z_B = Z_{22} - Z_{21}$$

$$Z_C = Z_{12} = Z_{21}$$

2) π -equivalent representation of a 2-port reciprocal n/w in terms of Y-parameters

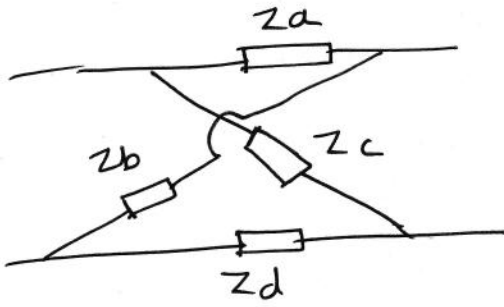


$$Y_A = Y_{11} + Y_{12}$$

$$Y_B = Y_{22} + Y_{21}$$

$$Y_C = -Y_{12} = -Y_{21}$$

* Lattice N/w



If $Z_a = Z_d$ and $Z_b = Z_c \rightarrow$ Symmetrical and balanced lattice

If $Z_a \neq Z_d$ (or) $Z_b \neq Z_c \rightarrow$ Asymmetrical and unbalanced lattice.

Attenuators $\rightarrow$ Fixed potentiometers

* Representation of Lattice n/w in terms of Z-Parameters:

$$Z_{11} = Z_{22} = \frac{Z_b + Z_a}{2}$$

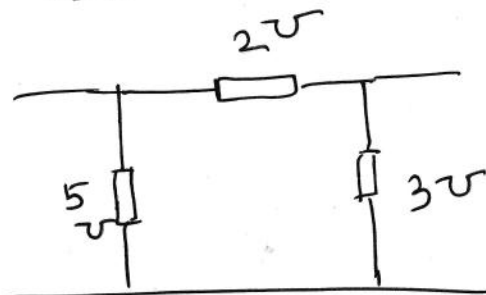
$$Z_{12} = Z_{21} = \frac{Z_b - Z_a}{2}$$

$$Z_{11} + Z_{12} = Z_b$$

$$Z_{11} - Z_{12} = Z_a$$

GATE:- The port currents of a 2 port n/w are given by,
 $I_1 = 7V_1 - 2V_2$ then its π equivalent is —?
 $I_2 = -2V_1 + 5V_2$

Sol:-



$$[Y] = \begin{bmatrix} 7 & -2 \\ -2 & 5 \end{bmatrix}$$

yes Reciprocal
 so, π possible.

Note:

$$[z] = [y]^{-1} \quad \& \quad [y] = [z]^{-1}$$

$$[h] = [g]^{-1} \quad \& \quad [g] = [h]^{-1}$$

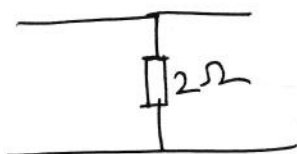
$$[T] \neq [T']^{-1} \quad \& \quad [T'] \neq [T]^{-1}$$

These are invertible only if n/w is reciprocal.

IES 2012

Det. $[z]$ & $[y]$.

→ None



Sol:

$$[z] = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \Omega$$

$$\det |z| = 0.$$

→ singular matrix

⇒ $[y]$ can't be found

$[y] \rightarrow$ not existing

* Inter conversion b/w '2' 2-port n/w s:

Q: Represent h-parameters in terms of y-parameters.

y

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

$\rightarrow V_2 = 0$

$$I_1 = Y_{11}V_1$$

$$I_2 = Y_{21}V_1$$

$\rightarrow I_1 = 0$

$$Y_{11}V_1 = -Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

h

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

$h_{11} \times h_{21} \rightarrow V_2 = 0$

$$h_{11} = \frac{1}{Y_{11}}$$

$$h_{21} = \frac{Y_{21}}{Y_{11}}$$

$h_{22} \times h_{12} \rightarrow I_1 = 0$

$$h_{22} = \frac{I_2}{V_2}$$

$$I_2 = Y_{21} \left(-\frac{Y_{12}}{Y_{11}} \right) V_2 + Y_{22}V_2$$

$$I_2 = \left(\frac{\Delta Y}{Y_{11}} \right) V_2$$

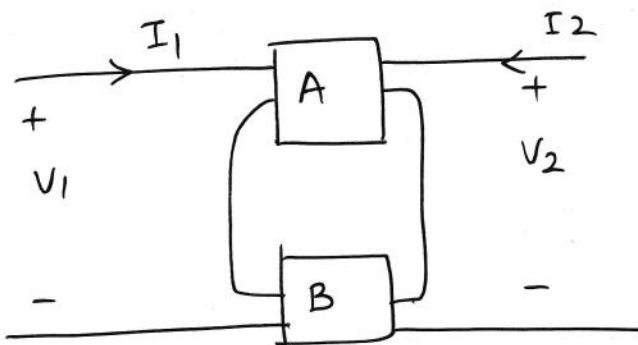
$$h_{22} = \frac{\Delta Y}{Y_{11}}$$

$$h_{12} = -\frac{Y_{12}}{Y_{11}}$$

$$[h] = \begin{pmatrix} 1/Y_{11} & -Y_{12}/Y_{11} \\ +Y_{21}/Y_{11} & \Delta Y/Y_{11} \end{pmatrix}$$

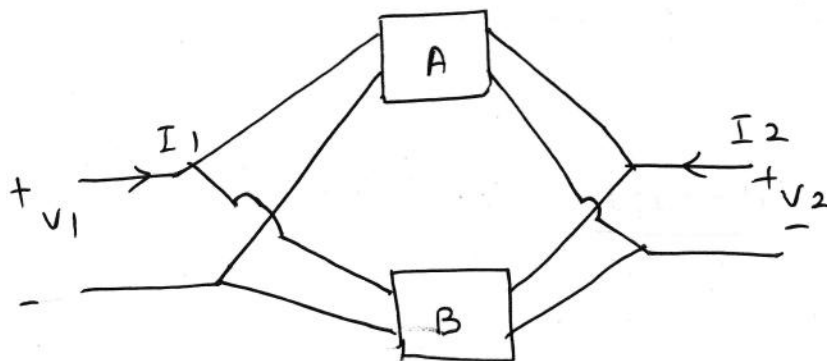
* Inter-connection b/w '2' 2-port n/w's:

1) Series-Series



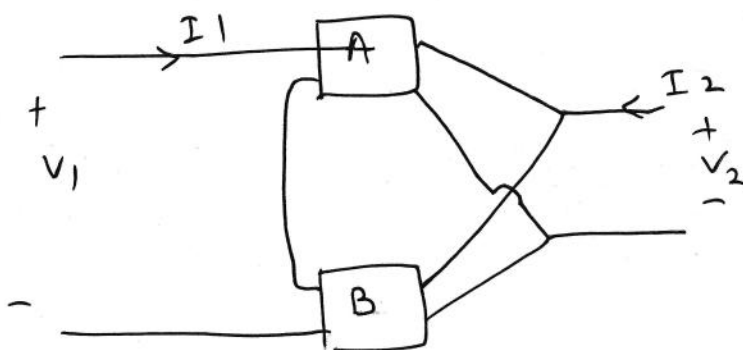
$$[Z_T] = [Z_A] + [Z_B]$$

2) Parallel-Parallel:



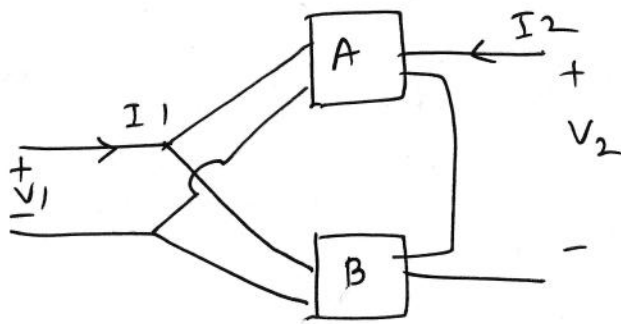
$$[Y_T] = [Y_A] + [Y_B]$$

3) Series-Parallel



$$[h_T] = [h_A] + [h_B]$$

4) Parallel-Series



$$[g_T] = [g_A] + [g_B]$$

5) Cascade / Chain:



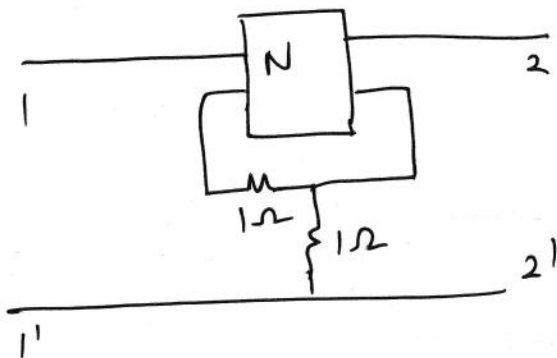
$$[T_T] = [T_A] \cdot [T_B]$$

↓
Multiplied.

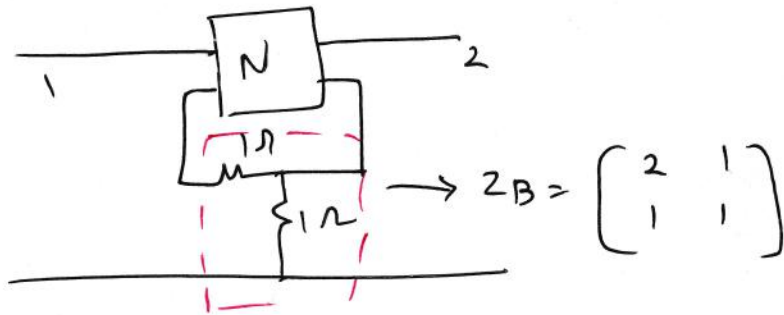
GATE

If z parameters of a n/w are $z_N = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

$[z]$ of total n/w is —.



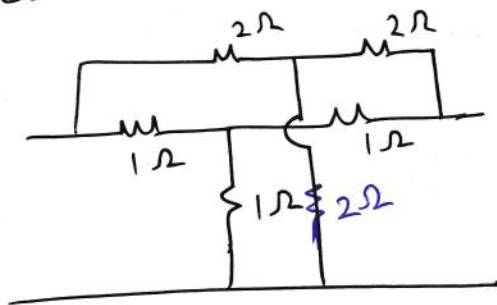
Sol:



$$Z = [Z_A] + [Z_B] = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 2 \end{bmatrix} \Omega$$

GATE:

Det. the 'Y' parameters of



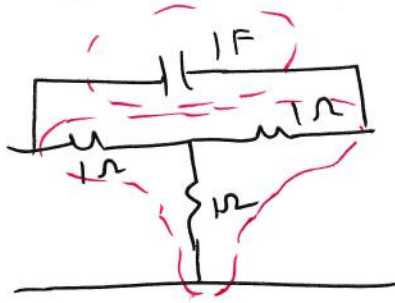
$$\begin{array}{c} 1\Omega \quad 2\Omega \\ \text{---} \text{---} \\ | \\ 1\Omega \end{array} \rightarrow Z_A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \rightarrow Y_A = \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix}$$

$$\begin{array}{c} 2 \quad 2 \\ \text{---} \text{---} \\ | \\ 2 \end{array} \rightarrow Z_B = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \rightarrow Y_B = \begin{bmatrix} 1/3 & -1/6 \\ -1/6 & 1/3 \end{bmatrix}$$

$$Y = \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1 \end{bmatrix} \text{ S}$$

GATE

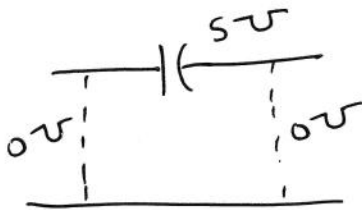
$[Y] =$ _____



Sol.



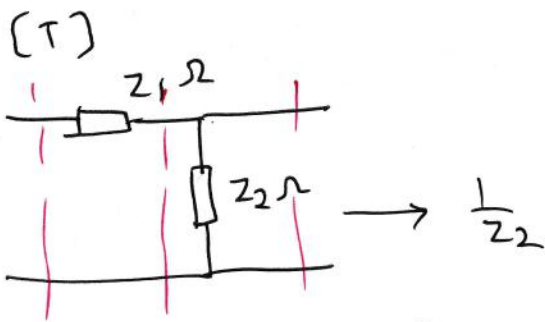
$$\rightarrow [Y_A] = \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix}$$



$$[Y_B] = \begin{bmatrix} s & -s \\ -s & s \end{bmatrix}$$

$$[Y_T] = \begin{bmatrix} 2/3 + s & -(1/3 + s) \\ -(1/3 + s) & 2/3 + s \end{bmatrix}$$

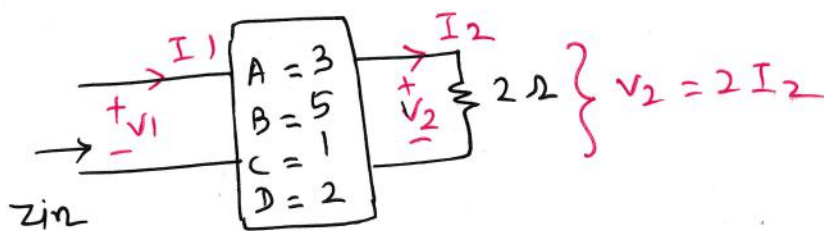
Q:



$$\begin{bmatrix} 1 & z_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{z_2} & 1 \end{bmatrix} = \begin{bmatrix} 1 + \frac{z_1}{z_2} & z_1 \\ \frac{1}{z_2} & 1 \end{bmatrix}$$

GATE/EE

Determine input impedance



$$z_{in} = \frac{V_1}{I_1} = \frac{3V_2 + 5I_2}{1 \cdot V_2 + 2I_2} = \frac{6I_2 + 5I_2}{2I_2 + 2I_2} = \frac{11}{4} \Omega$$

IES/14

The transmission parameters of a 2-port n/w consisting of only impedances is $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$.

IES/14

If all the impedances in the n/w is doubled, then the new transmission parameters.

Sol.



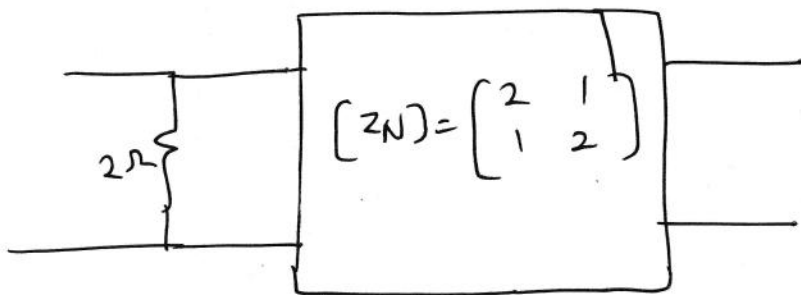
$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$



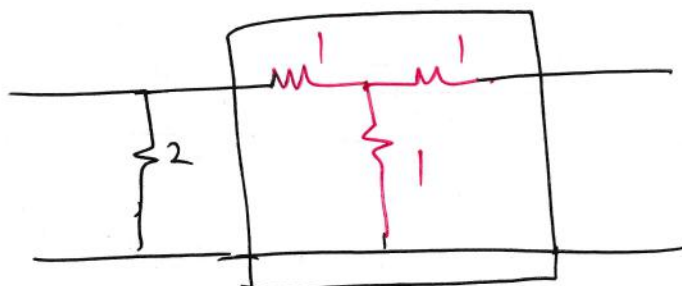
$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 1/2 & 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} A & 2B \\ C/2 & D \end{bmatrix}$$

GATE Determine (T)

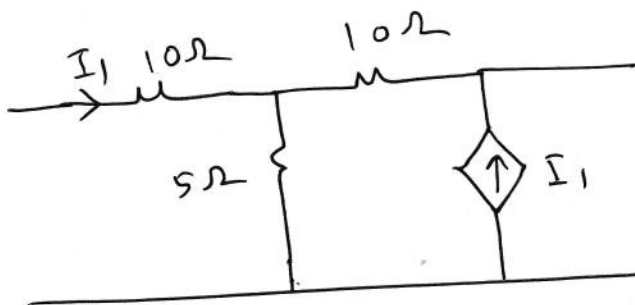


Sol.



$$\begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} =$$

IES(c) Det. Z parameters for



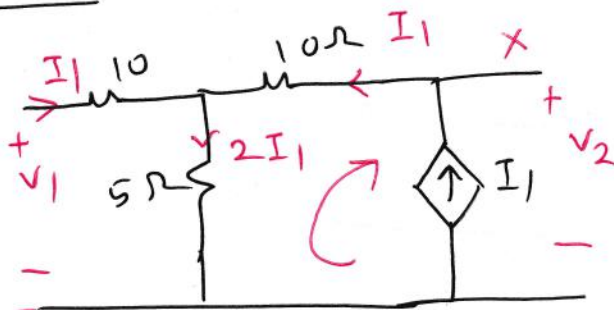
Sol.

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

$$Z_{11} = \frac{V_1}{I_1} / I_2 = 0$$

$$Z_{21} = \frac{V_2}{I_1} / I_2 = 0$$

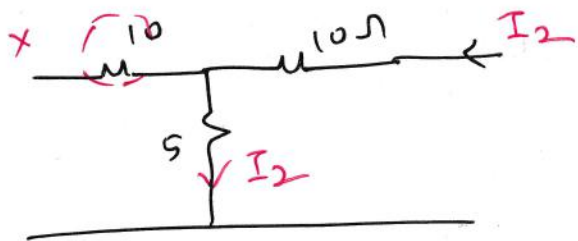


$$V_1 = 10 I_1 + 10 I_1$$

$$V_1 = 20 I_1 \Rightarrow Z_{11} = 20 \Omega$$

$$-(10 I_1 + 10 I_1) + V_2 = 0$$

$$V_2 = 20 I_1 \quad Z_{21} = 20 \Omega$$



$$V_2 = 15 I_2$$

$$Z_{22} = 15\ \Omega.$$

$$Z_{12} = \frac{V_1}{I_2} = \frac{5 I_2}{I_2} = 5$$

$$\underline{\underline{\begin{pmatrix} 20 & 5 \\ 20 & 15 \end{pmatrix} \Omega}}$$